

PAVAN KUMAR ETTA

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EDUCATION

University of Massachusetts, Dartmouth | GPA: 4.0

M.S. in Computer Information Science, Data Science

North Dartmouth, USA

Aug 2024 - May 2026

Indian Institute of Information Technology | GPA: 3.2

B.Tech in Electronics and Communication Engineering

SriCity, INDIA

Aug 2017 - Jun 2021

SKILLS

Programming: Python, JavaScript, TypeScript, C/C++, Java, Go

Software Engineering: Data Structures & Algorithms, OOP, Microservices, Distributed Systems

Backend & APIs: Django, Flask, FastAPI, REST APIs, gRPC, GraphQL, Event-Driven Architecture

Frameworks: jQuery, Angular, Node.js, Vue.js, Next.js, Flutter, Nginx, Figma, HTTPS

Databases & Caching: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch

Cloud & Infra: AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless

Data Engineering: Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt

DevOps & Observability: Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI

WORK EXPERIENCE

Graduate Research Assistant | University of Massachusetts, Dartmouth | USA

Mar 2025 – Dec 2025

- Led development of a financial-reporting platform using **Apache Spark**, **Hive**, and **Hadoop** to process uploaded documents, migrated report-generation logic from Java to Scala, and built a React.js validation interface for verifying NLP-generated reports.
- Integrated LangChain for AI workflow orchestration and optimized ChatGPT-based text extraction, improving extraction accuracy by **30%**, implemented BERT-based classification with **93%** accuracy.
- Designed and implemented a scalable backend pipeline using **Flask**, **MongoDB**, **Redis**, and **Docker** to orchestrate document uploads, asynchronous report generation, validation, and persistent storage, improving system reliability and maintainability.

Software Engineer II | Rhenix Lifesciences LLP | India

Aug 2021 – Aug 2024

- Engineered CureOrg with integrations across 5+ wearable platforms using **OAuth 2.0** and REST APIs, enabling synchronization of health data for **100+ users** and processing **1M+ health records**, including activity, sleep, and heart-rate metrics.
- Automated daily health-data processing with Python jobs and **Django** management commands on **AWS EC2**, reducing manual collection by **90%** through token refresh, JSON parsing, retries, validation, and MySQL persistence.
- Improved reliability of cross-functional health-data services using **Kubernetes** orchestration and optimized **GraphQL** queries, while implementing audit logs and alerts that reduced synchronization debugging time by **40%** and accelerated failure detection.
- Built a scalable data-access layer using indexed MySQL queries, **Redis caching**, and paginated **REST endpoints**, improving health-metrics retrieval performance by **70%** for frequently accessed user data.
- Established unit, integration, and regression testing for Go and Python services, validating **PostgreSQL** workflows and Dockerized deployments across **ECS**, **S3**, and **SQS** while improving defect detection, maintainability, and release confidence.
- Owned cross-platform feature delivery across backend, Android, iOS, QA, and DevOps teams by translating customer requirements into technical tasks, resolving release blockers, and supporting Google Play and App Store deployments.

Software Engineer Intern | Cure Science Institute | USA

Jun 2021 – Aug 2021

- Developed a **BLE-based** asset tracking system using an Android app, BLE beacons, gateways, and REST APIs, enabling real-time location data uploads to AWS S3 with persistent background services for **24/7** tracking and improved battery efficiency.
- Implemented **Go-based** monitoring services with health checks, structured logging, and configuring **New Relic** alerts and Slack notifications to detect offline gateways, API failures, and delayed updates, improving production visibility and incident response.
- Engineered a scalable backend using **AWS Lambda**, API Gateway, DynamoDB, RDS, and scheduled jobs to process device telemetry, persist location data, and support real-time asset monitoring and analytics.

PROJECTS

Distributed API Gateway & Rate Limiting Service [\[GITHUB\]](#)

(Python, Go, AWS, Redis, Docker, Kubernetes)

- Designed a cloud-native API gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

Fault-Tolerant Sharded Key-Value Store [\[GITHUB\]](#)

(C++, Raft, SQLite)

- Engineered a fault-tolerant sharded key-value store in **C++** using Raft consensus, SQLite, leader election, log replication, and replicated state machines to maintain consistency and support recovery from server failures.

Automated Subjective Answer Evaluation System [\[GITHUB\]](#)

(Flask, BERT, Machine Learning, Docker, AWS EC2/S3)

- Developed an automated answer-evaluation platform using Python, Flask APIs, BERT, Decision Trees, Random Forest, and XGBoost to process more than 10,000 responses and achieve **95%** classification accuracy.

Wandermate [\[GITHUB\]](#)

(React, TypeScript, Node.js, MongoDB, Socket.IO)

- Led the development of a full-stack travel platform using TypeScript, Node.js, and MongoDB, integrating Socket.IO for real-time messaging, PropelAuth for secure authentication, Mapbox for interactive maps, and reusable APIs for third-party integrations.